Clark University

Running a Non-Profit in a Recession: Analysis of Countercyclical Financial Practices

Economics Honors Thesis

by

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Abstract

Non-profit organizations suffer from an overall cyclical 'dilemma' during an economic recession: the demand for philanthropic services increases (e.g., more people will request support for food services) meanwhile non-profit revenues decreases (e.g., people have less disposable income to donate). This paper seeks to define 'resilient financial practices' for the US non-profit sector to follow that are correlated with helping non-profits to do more when the economy does worse. This paper utilizes Ordinary Least Squares (OLS) panel data regressions with fixed effects (year, state, industry, and firm), alongside a probit regression model on 2.88 million observations of 501(c)(3) tax exempt organizations from 2009-2023. The results confirm the non-profit sector remains largely procyclical. However, maintaining highly liquid, unrestricted revenue streams or pre-existing pipelines for government grants is correlated with increases in an organization's probability to expand when society needs them most. For future research, this paper recommends testing pre-defined financial practices from the literature, examining specific balance sheet reallocations during shocks, pairing quantitative data with qualitative interviews to determine if countercyclical planning is an active concern among non-profit business leaders.

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1. Introduction

Contrary to publicly traded for-profit companies, non-profits do not need to seek continual expansion; rather, they respond to changing demands and needs of the community they volunteer to support. Theoretically then, many non-profits do not need to expand when the broader economy is performing well – because their constituents will be better off and in less need for philanthropy. However, in an economic downturn or recession, demand for nonprofit services inherently increases, suggesting that nonprofits should counter-cyclically expand their program offerings to match this heightened need. Despite this expectation of altruistic countercyclicality, many non-profits are constrained by their budget and available revenues, which follow a procyclical cycle. In a recession, investment income crashes, states shrink their budgets, and individual giving falls. But not all revenue sources for a given organization fall in a recession – some organizations may save surpluses from good times for bad times, the federal government may pass a non-profit stimulus package, and organizations can reshuffle assets.

This creates a puzzle: What revenue sources are most correlated with helping non-profits do more when the economy is doing worse? To answer this, I conduct a longitudinal panel data regression analysis utilizing digital Form 990 tax returns from tax-exempt organizations across the United States. The final dataset comprises nearly 2.88 million observations from 391,529 unique organizations spanning a 15-year period from 2009 to 2023. Since non-profits lack a simple "bottom line," I use program expenses as a proxy for the demand-side output of mission fulfillment, and total revenues to represent the supply-side input of peak capacity. I use an Ordinary Least Squares (OLS) model with year, state, industry, and firm fixed effects to observe the correlations between financial behaviors and economic conditions.

The findings confirm that the non-profit sector is largely procyclical, aligning with recent literature. The firm fixed-effects model reveals that for a one percentage point increase in real state GDP, a non-profit's total revenue and program expenses both increase by approximately 0.6%. However, when holding budget, or an organizations given revenue, as constant, the economic conditions of a non-profit's state no longer have a significant effect. Instead, the change in program expenses is mostly explained by changes in total revenue. The revenue-controlled model reveals that for a one percent increase in a non-profit's total revenue, program expenses increase by 0.82%. However, countercyclical behavior is still observable across some industries of organizations and different revenue sources. Furthermore, when testing against the specific macroeconomic shock of the Covid-19 recession, the models reveal that countercyclical behavior probability is significantly correlated with an organization's revenue streams.

While existing literature has confirmed the procyclical trend of larger non-profit enterprises and the increased demand by constituents as the economy declines (Exley, Lehr and Terry 2023); (Bridgeland, et al. 2009), these studies have historically been limited by a narrow focus on individual recessions. Literature investigating optimal non-profit practices is limited to case studies of specific industries or revenue models (Altamimi and Liu 2022). This thesis contributes to the existing body of literature by observing multiple industries, geographies, and organizations across a large timeframe to find resilient practices across the entire non-profit enterprise in the United States.

1. Literature Review

Research into non-profit economics was limited for a long time but has been continually growing in volume in the past few decades, moving from abstract theory to empirical study of financial practices. Hansmann (1980) wrote some of the earliest and most influential theoretical work on the role of nonprofits, establishing the "nondistribution constraint" that distinguishes non-profits from for-profit firms. Although not technically barred from making a profit, a non-profit corporation can only use the surplus for reinvesting its capital to advance the non-profit's stated mission. This difference distinguishes non-profit operations because it should vastly improve trust among consumers and encourage the expansion of nonprofit enterprises across various sectors of the economy (Hansmann 1980). Although more concerned with the legal perspective than economics and theory, Hansmann's research is foundational and expanded upon, including in economics. Hansmann's theoretical argument can establish why non-profits may operate outside of typical, cyclical trends and require unique financial procedures.

The transition toward a formal economic perspective of non-profit research accelerated after work from Rose-Ackermann (1996), who directly confronted economists' assumptions that non-profits were "noneconomic institutions." Using empirical methods from economics and psychology, Rose-Ackermann (1996) supports the economics logics for altruism, nonprofits' role in solving information asymmetries, and the high trust perceived in non-profit corporations. Her research is important because it was some of the earliest encouragement for using regression analysis to analyze non-profit behavior over time. However, while Rose-Ackermann provided the early map for empirical study, her work lacked large-scale panel data to see how organizations behave over time. My study fills this gap by applying these economics logics to a longitudinal dataset of almost 2.9 million observations.

Despite the theoretical potential for non-profits to be countercyclical, the work by Exley, Lehr, and Terry (2023) quantified a problematic procyclical trend by analyzing the Covid-19 recession. Their research found that during an economic downturn, non-profit organizations commonly shrink their financial size. Revenues, expenses, and balance sheets are all strongly procyclical, disappointing public expectations for certain industries to expand countercyclically, such as those providing housing and food assistance. Similarly, researchers Bridgeland, McNaught, Reed, and Dunkelman (2010) focused on what they termed the "quiet crisis" of shrinking non-profit sector following the 2008 recession. They find equivalent results: organizations faced diminishing charitable donations, reduced state and local government budgets, and increasing human need, which resulted in large non-profit budget cuts, layoffs, and halted programs, especially across human service and cultural organization industries. While these studies identify the problem during specific macroeconomic shocks, their shortcoming is this narrow focus on an individual recession. My research expands on their work by looking at a 14-year period (2009–2023) to see if this procyclicality is an inherent trait of non-profit enterprise or just an observable reaction to a recession.

Building on the observation that nonprofits are highly vulnerable to economic downturns, Arbogust (2020) compounds the results of Exley et al. (2023) and Bridgeland et al. (2010) by conducting a longitudinal analysis of specific financial practices over time that can predict future failure of non-profit organizations. Significant changes in revenue concentration, administrative costs, or new unrelated business income are the main statistical differences between the organizations that fail versus ones that continue to succeed (Arbogust, 2020). The vulnerabilities identified in these studies motivate my research into financial practices that improve

countercyclical resilience. I seek to use a similar method as Arbogust by using a probit regression model to find practices correlated with ‘successes’ of a non-profit.

Recent research into optimal financial practices has begun to challenge ‘conventional wisdom’ by identifying how specific management choices can improve resilience. For example, Altamimi and Liu's (2021) research into the optimal overhead ratio challenges the assumption to ‘keep it as low as possible’ by demonstrating an inverted U-shaped relationship where increased overhead spending improves program outcomes, up to an optimal point. Further, Jones, Kitching, Roberts, and Smith's (2012) research about how well-managed non-profits can ‘weather the storm’ by saving revenues demonstrates that charities can actively smooth their spending over good and bad economic times by saving a substantial portion of revenue increases and then intentionally using those savings to mitigate future revenue reductions when the economy worsens. However, this type of research is traditionally limited to a ‘case study’ or specific group of organizations (i.e. arts/culture non-profit industry or non-profits that rely on independent donations as their revenue source) and so results cannot be confidently generalized to firms outside of this scope. I hope to address this shortcoming by analyzing multiple industries, geographies, and types of organizations to find financial practices that are consistent across these large groups.

Also following a broad scope of organizations, specifically all non-profits following the Italian economic crisis in 2008, Pollifroni, Militaru, and Socaciu (2014) establish economic theory to support non-profit countercyclicality. They argue that countercyclicality arises when organizations combine the nondistribution constraint’s focus on durability with a commitment to corporate social responsibility and find specific financial practices correlated with improving a non-profit’s ability to be countercyclical. They build on early theoretical research and find

empirical support that the “nondistribution constraint” (see Hansmann, 1980) of non-profits forces them to reinvest capital into the long-term durability, fulfilling the first condition. A non-profit’s voluntary goal to maximize public utility, run by unselfish stakeholders, and lack of proprietary rights fulfill the second condition for corporate social responsibility. Important to the real observable outcomes, both of these conditions are necessary but not sufficient for countercyclicality. A non-profit must still actively pursue the ethical managerial style and intangible remunerations (i.e., employee benefits, remote work, and organizational flexibility) to improve countercyclicality (Pollifroni, Militaru and Socaciu 2014). Their study provides a strong theoretical framework and support but is limited to the Italian context, which has different national policies, culture, and business practices compared to the United States. Nonetheless, I seek a similar test of whether these same countercyclical outcomes can be observed across the diverse landscape of the United States non-profit sector.

Effectively measuring when non-profits are able to ‘do more’ in a recession for my thesis requires a well-supported methodology, because non-profits lack a simple "bottom line" like for-profit businesses to measure. It is impossible to choose one financial variable to generalize all non-profit success. However, Kim (2017) provides a strong justification for using program expenses as a proxy for effectiveness, finding that program spending is closely correlated with actual program outcomes like attendance or registrations.

Furthering this, research from Ma and Beaton (2024) shows a non-profit is better able to respond to economic shocks if they are more successful at a baseline level, when comparing to peers/competitors (Ma and Beaton 2024). So, I use the financial variable program expenses function as a proxy for and mission success, reflecting the organization's active effort to meet community needs in changing economic conditions.

Finally, to control for the unobservable traits of organizations across the country, I follow the recommendations by Ba, Berrett, and Coupet (2023), who describe a high-level approach to conducting panel data research for non-profit studies. They encourage using a fixed effects OLS regression model as it is best to control for unobservable traits of organizations (e.g., leadership styles, values) and larger scale effects (e.g., market structures) that will bias results. By using their recommended methodological approaches on a dataset significantly larger than previous studies, I hope to identify how non-profits respond to the economic cycle and what financial practices might be most correlated with resiliency during a recession or countercyclicality.

2. Data Measurement and Exploration

3.1 Collection

This thesis looks to explore cyclical trends of financial practices for non-profit organizations in the United States. For this, the dataset is made up of digital copies of Form 990 documents, which describe financial details and business operations of filers, sent by every tax-exempt organization in the United States each year from 2009 – 2023, to the Internal Revenue Service (IRS). Per IRS requirements for filing Form 990: only businesses filling for income tax exemption under section 501(c)(3) of the Internal Revenue Code, based in any of the 50 US states, with gross receipts of greater than $200,000 in one year or net assets of greater than $500,000 are included in the dataset. In addition, I add further restrictions to this dataset, explained below.

Digitally summarized copies of yearly Form 990s are published online by the Internal Revenue Services (IRS) as .XML file formats (IRS.gov 2019). The National Center for Charitable Statistics (NCCS), a part of the non-profit and philanthropy research department of the Urban Institute, converts this raw data into data tables organized by year and header sections

of the Form 990 document (Lecy 2023). I compiled this collection from the NCCS of .CSV files into a raw dataset, then added supplemental data, then imposed necessary restrictions.

This preliminary dataset of Form 990 data is in panel form. Each organization fulfilling the IRS's requirements is observed in at least one year of data. Each organization is only observed once per year; however, not all the same entities are included in each year. This makes the panel data unbalanced. The data reports raw variables for revenues, expenses, activities, and governance of each organization. This includes granular financial data for the entire scope of operations (i.e., "Grants and other monetary assistance to domestic individuals," Part IX: Statement of Functional Expenses), categorical information of the entity (i.e., "Form of Organization," Part 0: Header), and summarized finances and balance sheet (i.e., "Total Assets," Part I: Summary). For this thesis, I rely on mostly summarized financial variables from the dataset.

To supplement the Form 990 dataset, the Internal Revenue Service collects Primary Activity Codes or National Taxonomy of Exempt Entities (NTEE) classification codes, but not both, for every single organization (IRS.gov 2017). NTEE codes are the newest format of non-profit business classification adopted by the IRS.

While filing a Form 990, or tax exemption form, an organization could list up to three activity codes to describe up to three of their main products/services provided. Exempt organizations founded before 1995 must continue using old activity codes. The activity codes for all non-profit organizations, past and present, are in the Internal Revenue Service's Business Master File (IRS.gov 2017). Organizations founded after 1995 must use the new NTEE codes. The NTEE code for all non-profit organizations, past and present, is also in the Business Master File. This master file is divided for each US state, based on organization headquarters location.

For each state, there is a .CSV file. First, I combined all 50 files, one from each state, then coded a matching function so that organizations filing an old activity code are replaced by the function to the nearest NTEE code. The IRS provides information describing the categories for both activity and NTEE codes, as well as examples of organizations belonging to each category. Using this information, I manually matched each activity code category to its nearest related NTEE code category.

Below is a table of all NTEE code categories:

Table 1. Distribution of Non-profit Industries

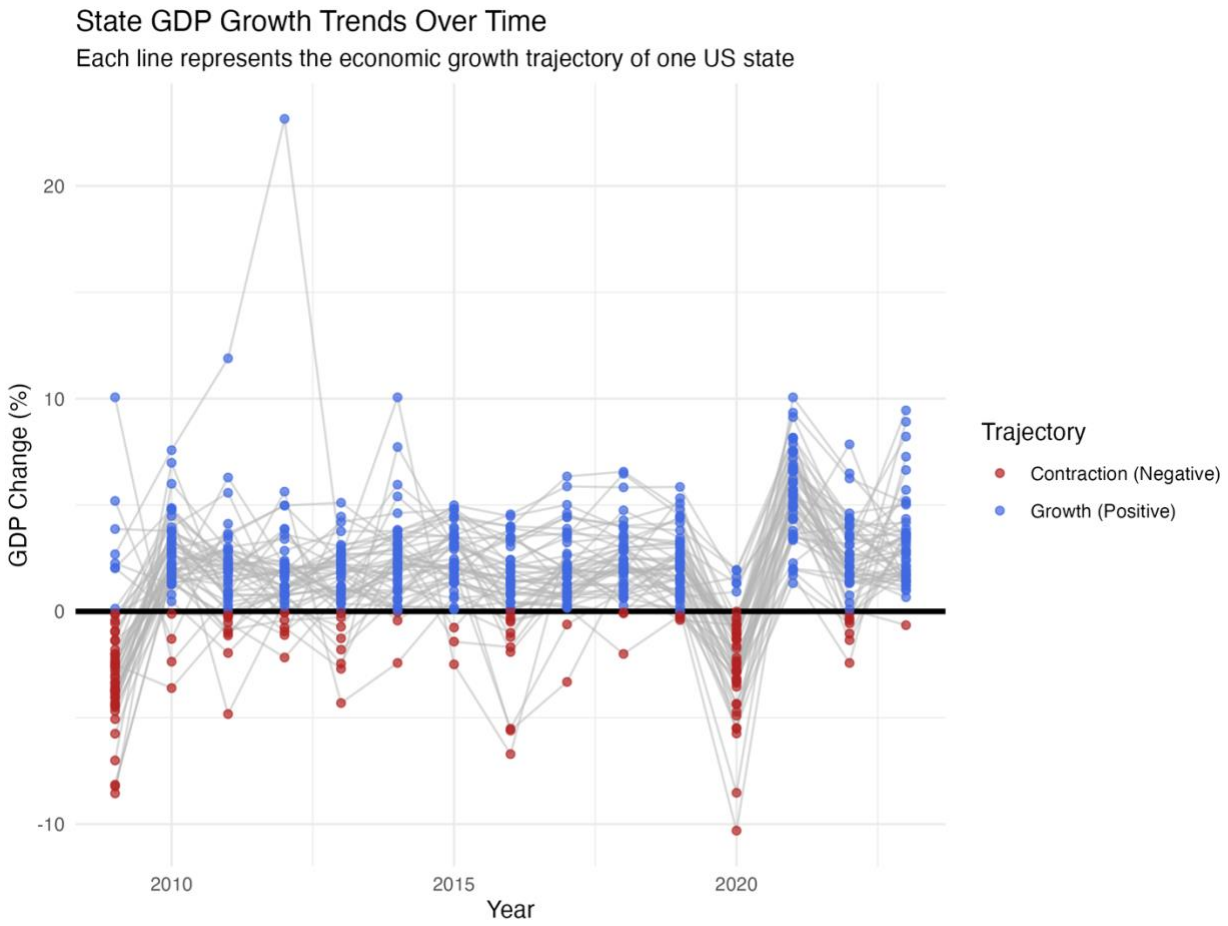
Letter	NTEE Code Industry
A	Arts, Culture & Humanities
B	Education
C	Environment
D	Animal-Related
E	Health Care
F	Mental Health & Crisis Intervention
G	Voluntary Health Associations & Medical Disciplines
H	Medical Research
I	Crime & Legal-Related
J	Employment
K	Food, Agriculture & Nutrition
L	Housing & Shelter
M	Public Safety
N	Recreation & Sports
O	Youth Development
P	Human Services
Q	International, Foreign Affairs
R	Civil Rights, Social Action & Advocacy
S	Community Improvement & Capacity Building
T	Philanthropy, Voluntarism & Grantmaking
U	Science & Technology
V	Social Science
W	Public & Societal Benefit
X	Religion-Related
Y	Mutual & Membership Benefit
Z	Unknown or Unclassified

To further supplement this dataset, I gathered real Gross Domestic Product (GDP) data for each state. The Bureau of Economic Analysis (BEA), an agency under the United States Department of Commerce, collects this data (BEA.gov 2026). The BEA collects and verifies this data for each American state and territory. For this thesis, I use only real GDP, after accounting for national inflation, for each state. All the data is in a .CSV file with yearly dollar values for each state from 1997 – 2024.

I subset this data to only the years 2009 – 2023 and convert the real GDP dollar value to a growth percentage. Since I have a GDP value for 2008, I have the complete range for my final dataset.

Figure 1 shows the overall state GDP growth rate trend in the final data set. There is a lot of variation across the states and over time. In the beginning of the data set, for 2009, a large number of states are still suffering from the 2008 Great Recession and have contracting GDP. In the years following until 2020, most states exhibit growth year over year, but with exceptions. The 2020 recession affected almost all 50 states, but almost all equally recovered the next year.

Figure 1. State GDP Growth, over time



These three data sources: Form 990 returns, Activity/NTEE code industries, and real GDP make up my final data set. No restrictions were placed on Activity/NTEE codes or real GDP data beyond sub setting the data to match my universe of non-profit observations. However, Form 990 returns include potential mistakes by filers, empty observations, and outliers so I must impose some data restrictions.

3.2 Restrictions

Firstly, according to 501(c)(3) status regulations, an organization that does not follow filing procedures for three consecutive years loses its tax-exempt status (IRS.gov 2017). Additionally, an organization that is not contributing a public benefit is at risk of losings its

special status. So, three consecutive years of zero program expenses, assuming this shows zero public benefit, is a reasonable restriction to make because these are organizations that may lose their non-profit status. This drops 1,470,587 observations down from 5,002,756.

Also, large institutions like hospitals and universities have significantly greater legal and financial guidance than smaller organizations and operate differently (Shon, Hamidullah and McDougle 2019). Shon et al. (2019) finds that changing revenue concentrations - the primary experimental variable in their research - does not have a significant effect on hospital or school spending patterns. Further, these institutions represent some of the largest outliers in my dataset and demand for their services is likely inelastic. Mortensen and Chen (2013) demonstrate that hospital visit rates are the same before and during a recession, supporting inelastic demand. For universities, demand for their services does increase during a recession like the other non-profit types, but so does total supply of the education, since enrollment increases accordingly (Schmidt 2018). Thus, universities are unique outliers amongst the overall trend of non-profit organizations and including them would skew the average behaviors for the rest of the Education industry. This drops 558,692 observations.

Next, I must drop observations filing with negative values for financial variables. Shon et al (2019) enforce a similar restriction in their research on non-profit financial health. It is possible some organizations incorrectly file their annual reports, creating negative values for variables such as total revenue or expenses and those filing errors should be removed. This drops 85,230 observations.

Fourth, I restrict my dataset to only organizations with a headquarters in the United States. My economic variables are based in the fifty states, and I do not have data for countries or states outside of this scope. Also, I hope for potential policy or business recommendations from

the results to be relevant for the universe of organizations and governments involved in the dataset and should exclude international organizations to remove biasing these recommendations. This drops 47,049 observations.

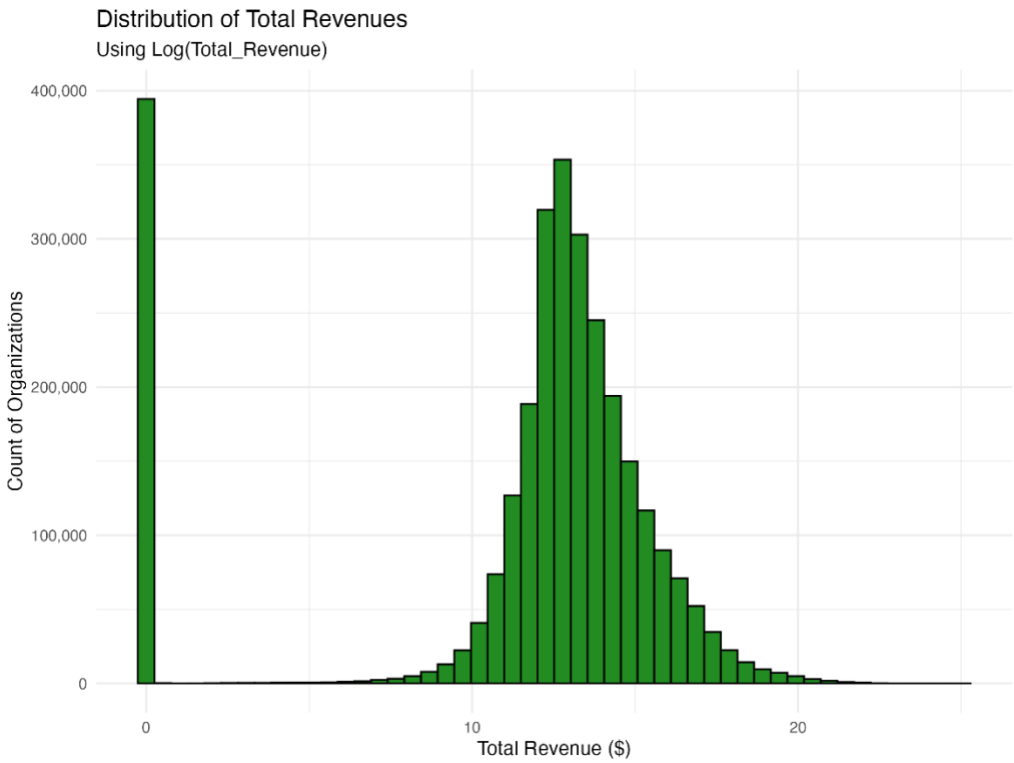
Lastly, I remove singletons from the data, so that every regression now uses the same sample size. Each organization must have at least two observations during my study period to be included in the analysis. This drops 63,501 observations.

From all these data restrictions, this limits the data set to 2,880,054 observations (from 5,002,756) across 391,529 unique organizations (from 738,790).

3.3 Description

Figure 2 shows the distribution of program revenues during the study period. Viewing the data with restrictions, the double skewing is no longer present because observing the log() observations successfully eliminate the effect of outliers. The data is mostly normally skewed, except with still a large concentration at zero. However, this is expected and certainly plausible when considering non-profit operating procedures.

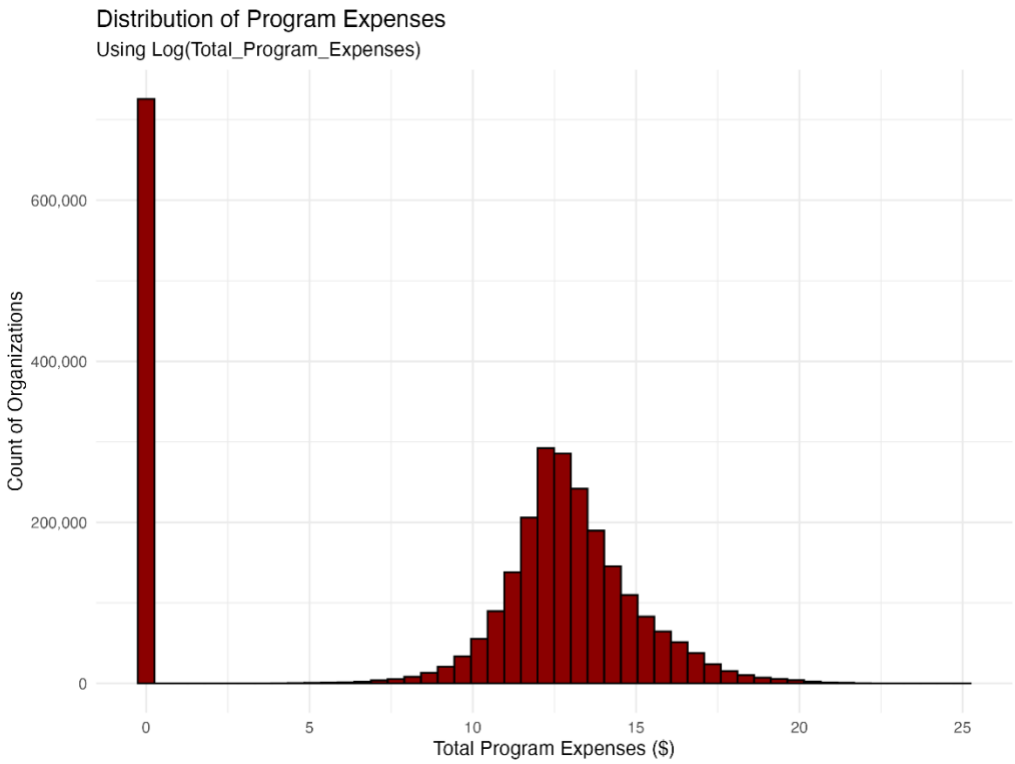
Figure 2. Distribution of Total Revenues



Firstly, many organizations experiencing financial difficulties may still be present in the dataset. The restriction is only for organizations that have no program expenses for three or more years, so this still includes organizations that may not earn a consistent revenue. Secondly, some organizations fund their operations via grants, which follow a cycle. An organization may receive a grant for a multi-year project, but must report all the revenue in year one, even if the grant is for two years of program expenses (Krebiehl 2019).

Figure 3 shows the distribution of program expenses. Again, many financially unhealthy non-profits may be unable to fund their operations and report zero program expenses. Regardless, organizations that choose to or must cease program expenses, without dissolving the organization entirely, are extremely valuable for this thesis, which is about financial health and macroeconomics of non-profit organizations.

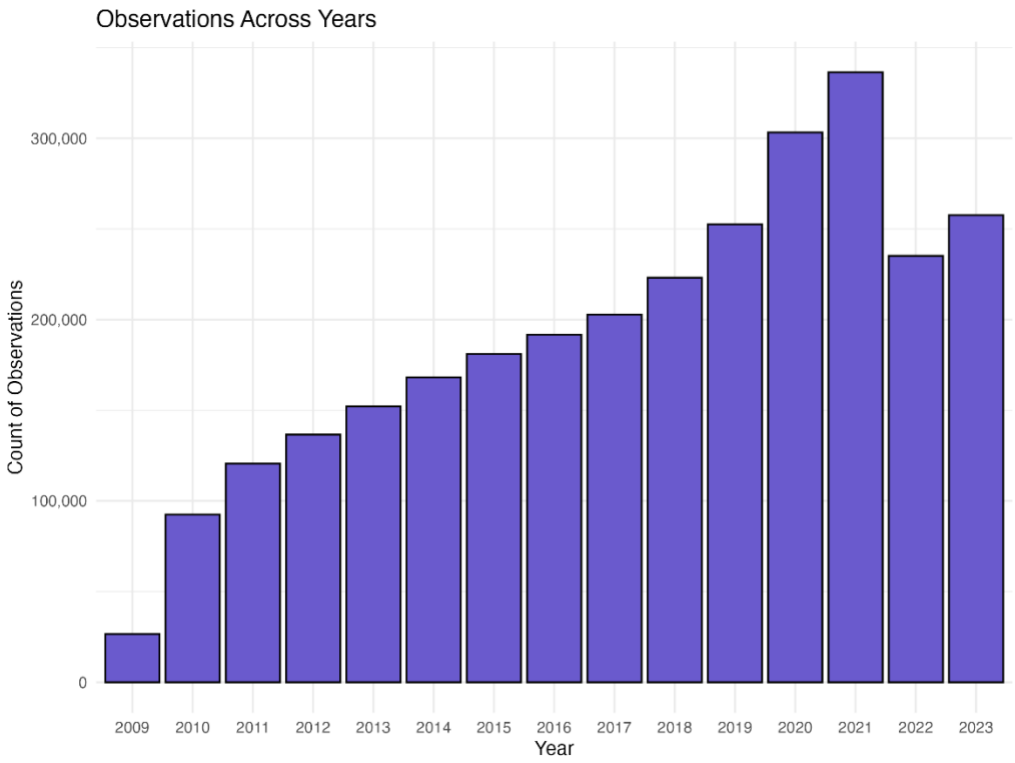
Figure 3. Distribution of Program Expenses



Another consideration, and potential bias for this thesis is non-profit organizations that operate without a budget. For example, organizations that rely on volunteers to mentor students, or business associations that connect people virtually. An organization with zero program expenses may or may not be an organization with zero social benefit however it is impossible to determine this from financial data alone. As such, I conservatively assume an organization with no program expenses is providing no social benefit.

Figure 4 shows the total number of non-profit organizations filing each year in the data set. The growth in the number of organizations is expected with rising wealth, country population, and a general rise of social entrepreneurship and non-profit funding following the recovery of the 2008 recession (Salamon and Toepler 2000). Jack and Patterson (2025) observe that the large growth in organizations since 2009 is primarily concentrated in the Human Services (NTEE industry P), Environment (C), and Education (B) industries, accordingly.

Figure 4. Number of Non-profits, by year



Most notably, there is a steep drop-off of over 50,000 organizations from 2021 to 2022. It is impossible to know why each organization fell out of the dataset, but there are many trends that occurred. Firstly, the ending of Covid-relief funds from governments, such as from the CARES Act which was a large cash stimulus targeted to the non-profit sector. Secondly, the IRS deadline was in 2022 requiring online tax filings. Many organizations reported difficulties in switching their tax filings system for the deadline, particularly among smaller, older organizations. Lastly, it could be the effect of the recession where revenue sources like donations dried up for non-profits and either shrunk to below the size requirement for Form 990 or dissolved entirely.

Across all variables, there is a wide range of observations and Table 2 shows a summary of these financial variables. All variables have organizations reporting zero expenses or revenues, which is expected from young and small non-profits or short-term business changes. On the

other hand, some non-profits, such as those working for the federal government or in research and development sectors may be receiving and spending massive amounts of money compared to their peers, over $11 billion in some years. Some of the largest organizations in the data set are receiving in revenues and spending in program expenses near $75 billion in given years. Amng the revenue sources, government revenue has the highest average, at $609,000. Among the entire dataset there is great variance of variables, with the standard deviations exceeding the averages for each financial variables.

Table 2. Descriptive Statistics for Selected Variables

Variable	Description	Observations	Mean	Standard Deviation	Minimum	Maximum
Donation Revenue	Revenue from campaigns and direct donations	2,880,054	$9,065.60	356,498.20	$0.00	$127,014,579.00
Fundraising Revenue	Revenue from fundraising events	2,880,054	$30,819.60	854,165.41	$0.00	$350,002,158.00
Government Revenue	Revenue from government grants	2,880,054	$609,078.65	15,940,399.11	$0.00	$11,069,708,938.00
Investment Revenue	Net income from investments	2,880,054	$40,498.89	1,465,053.77	$0.00	$717,215,986.00
Membership Revenue	Revenue from membership dues	2,880,054	$44,228.46	875,560.37	$0.00	$163,320,095.00
Other Revenue	Miscellaneous revenue from other sources (i.e., sale of property, etc.)	2,880,054	$141,399.33	4,528,116.07	$0.00	$2,366,107,467.00
Total Program Expenses	Total expenses directed towards programs. Represents the demand for the organization's services.	2,880,054	$6,636,765.48	150,719,949.06	$0.00	$71,256,037,114.00
Total Revenue	Total revenue from all sources. Represents the supply of the organization's services.	2,880,054	$9,090,684.18	174,916,735.29	$0.00	$75,101,306,911.00

4. Methodology

4.1 Independent Variables

My main explanatory variable is the percent change in each state's real gross domestic product (GDP) because it is a common metric to study economic conditions. This variable is adjusted for inflation, so fluctuations reflect real economic changes. I use state-level data rather than national GDP, so it captures the unique economic conditions that most impact the recipients and donors of a given non-profit.

4.2 Dependent Variables

Citizens have strong preferences for non-profits to expand services, especially in economically tough times (Exley, Lehr and Terry 2023). For my research to best capture this preference through the data, I capture the change in program expenses year-to-year. So, my first key dependent variable is "Total Program Expenses" because it measures the total money spent on activities advancing each non-profits mission statement. Other research may use more direct survey data to measure 'direct impact' towards the mission in a recession (Altamimi and Liu 2022), however I do not have the opportunity to survey for such data and use program expenses as a proxy for impact. Thus, for the market of a given non-profit's goods/services, program expenses represent the demand side of this function.

My second key dependent variable is "Total Revenues" because it measures the total supply of cash and other incomes available to a given non-profit each year. While program expenses reflect the intent to meet recipients or economic needs, total revenues represent the supply side of this function. It is the actual financial capacity of the organization to sustain or

expand its programs. By accounting for revenues, I can determine the change in program expenses explainable by budget constraints rather than responses to economic conditions.

4.3 Regression Models

I start with a pooled ordinary least squares (OLS) model (DeHaan 2021):

$$\mathrm{Y}_i = \beta_0 + \beta_1 X_i + \varepsilon_i \tag{1}$$

Where Y_i is the dependent variable or outcome for organization i (log total revenue). β_0 is the intercept of the equation when the effect is equal to zero. β_1 is the slope or magnitude of the effect of the key explanatory variable X (GDP Change %) on the dependent variable Y. However, may model suffer from omitted variable bias (OVB) from changes in conditions across time and space that are independent from a given organization's internal spend-save decisions about the economy. So, to improve the model, I include year fixed effects to account for secular time trends and any policies or shocks that are common across all organizations. In addition, I include state fixed effects to account for the fact that organizations across different states can face quite different economic environments. A non-profit operating in a highly regulated state may behave differently than one with a different set of conditions (Salamon and Toepler 2000).

To improve this model further, Mauri and Max (1998) find that it would be useful to control for the industry of companies and at the individual, firm level. This can help refine the results because companies have similar strategies to competitors. Further, each company has unique set of resources, even in the same industry, creating a unique effect of firm financial choices for this thesis. Holding firms as a constant would absorb most if not all industry fixed effects, since each firm only belongs to one industry, but the comparison between the two models will be useful in comparison and heterogeneity tests.

Adding industry fixed effects controls for unobserved, time-invariant characteristics that are unique to specific non-profit sectors creates the following revenue-side model.

$$\log(Total\ Revenues)_{year,state,industry} = \beta_0 + \beta_1(GDP\ Change\ \%)_{year,state} + \gamma_{year} + \alpha_{state} + \tau_{industry} + \varepsilon \quad (2)$$

In an alternative specification, I control for firm-specific fixed effects instead of industry and state fixed effects because even within the same state and industry, individual non-profit organizations have unique internal cultures, donor networks, reputations, and financial conditions.

$$\log(Total\ Revenues)_{year,firm} = \beta_0 + \beta_1(GDP\ Change\ \%)_{year,state} + \gamma_{year} + \omega_{firm} + \varepsilon \quad (3)$$

Where ω_{firm} is the firm fixed effect, capturing all unobserved, constant traits of the individual organization. Now, since a specific non-profit rarely changes its state headquarters location or industry classification, introducing firm fixed effects mathematically absorbs both the state (α_{state}) and industry ($\tau_{industry}$) fixed effects. So, ω_{firm} is highly effective to control for geographic, industry, and firm-specific variations simultaneously. For these reasons, this model acts as the definitive version of my supply-side regressions. While the main analysis examines the effect of GDP percentage change on total revenue, I further examine the effect of changes in GDP on sources of revenue, using $\log(Donations)$, $\log(Fundraising\ Events)$, $\log(Government\ Grants)$, $\log(Membership\ Fees)$, $\log(Investments)$, and $\log(Miscellaneous)$ as additional dependent variables.

In addition, I regress with program expenses as the dependent variable. Available revenues fundamentally constrain an organization's ability to spend on programs. As a result, I also control for budget in each year. Else, a non-profit organization might appear to counter

cyclically expand to fulfill increased demand when it is purely due to a large grant that must be used up.

$$\log(Program\ Expenses)_{year,firm} = \beta_0 + \beta_1(GDP\ Change\ \%)_{year,state} + \beta_2 \log(Total\ Revenue)_{year} + \gamma_{year} + \omega_{firm} + \varepsilon \quad (6)$$

Where $\beta_2 \log(Total\ Revenue)_{year}$ represents the total supply of cash and income available to the non-profit in a given year, it holds a given budget constant. Interpreting this model will be slightly different. Regressing $\log(x)$ on $\log(y)$ is interpreting the elasticity. So, a 1% increase in total revenue will be correlated with a β% change in program expenses. This is useful because by holding the supply of funds constant, I can accurately observe shifts in the demand side and its effect on non-profit activities.

Further, I test how GDP affects program expenses depending on the type of revenue, creates the following regression interaction models:

$$\log(\text{Program Expenses})_{year,firm} = \beta_0 + \beta_1(GDP\ Change\ \%)_{year,state} + \beta_2(Source)_{year,firm} + \beta_3(GDP\ Change\ \%)_{year,state} * \log(Source)_{year,firm} + \gamma_{year} + \omega_{firm} + \varepsilon$$

Where Source stands for Donations, Fundraising, Government Grants, or Membership Fees.

This is useful because it considers the effect of macroeconomic condition types of revenue and an organization's program expenses but with further analysis. Using the interaction term can help determine the specific effect of non-profit revenue forms in its ability to change program expenses during economic conditions. Controlling for revenue forms isolates the effect to only each form, which helps isolates the relationship between specific financial practices and countercyclicality.

Interpreting this model, a statistically significant, negative β_3 coefficient indicates a countercyclical driver. For example, as GDP decreases, an organization with a higher concentration of Government Grants will uniquely increase its program expenses.

$$(\text{COVID Expansion})_{year,state,industry} = \beta_0 + \beta_1 \log(Donations)_{year,state,industry} + \beta_2 \log(Fundraising\ Events)_{year,state,industry} + \beta_3 \log(Government\ Grants)_{year,state,industry} + \beta_4 \log(Membership\ Fees)_{year,state,industry} + \beta_5 \log(Investments)_{year,state,industry} + \beta_6 \log(Miscellaneous)_{year,state,industry} + \gamma_{year} + \alpha_{state} + \tau_{industry} + \varepsilon \quad (9)$$

Where (COVID Expansion) is a Boolean value created for each organization to represent whether they spend more money on program expenses in year 2020 compared to 2019. Since each organization is either a true or false, I cannot use firm fixed effects because there is no variation in the dependent variable across the same firm. So, I include γ_{year}, α_{state}, and $\tau_{industry}$ fixed effects as control variables.

Interpreting this regression, my explanatory variables are financial variables taken at the log level, and my dependent variable is true or false. So, for a percentage point increase in a specific revenue source, will be correlated with a percentage increase in the probability of expanding programs (COVID Expansion).

5. Results

First, I compare the baseline model while adding each fixed effect to see which has the greatest statistical significance and explanatory power. Each model adds new control variables and Table 3 shows those differences.

Table 3. The Effect of changes in state GDP on program expenses

	(1)	(2)	(3)	(4)	(5)
GDP Change %	-0.072***	-0.022***	-0.007**	-0.005*	0.006***
	(0.001)	(0.002)	(0.002)	(0.002)	(0.001)
Num. Obs.	2,880,054	2,880,054	2,880,054	2,880,054	2,880,054
R2	0.001	0.041	0.043	0.154	0.847
FE: year		X	X	X	X
FE: state			X	X	X
FE: industry				X	X
FE: firm					X

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: Regressing a non-profit's program expenses on state real GDP percentage change from prior year to current year, with each current year representing a Form 990 annual filing for tax-exempt status.

Column 1, the baseline demand-side results, representing a baseline Ordinary Least Squares (OLS) regression model, suggests a countercyclical relationship. For a one percentage point increase in real GDP, non-profit program expenses decrease by 7.2%. This result is significant at the 99% confidence level ($p < 0.01$).

However, as predicted, this model suffers from omitted variable bias (OVB). The coefficient (β_1) shrinks with each iterative model while the explanatory power (R^2) increases, and yet the statistical significance (p-value) remains high. Specifically, adding controls for years in column 2, states in column 3, and industry fixed effects in column 4 reduces the average size of the coefficient: column 4 shows that for a one percentage point increase in real GDP, non-profit program expenses decrease by 0.5%. This model is still significant at the 95% confidence level ($p < 0.05$).

A pivotal shift occurs in column 5, when adding firm fixed effects as the final degree of control in the model. The estimated effect becomes a positive sign, which indicates a procyclical trend. For a one percentage point increase in real GDP, non-profit program expenses increase by 0.6%, and this result is significant at the 99% confidence level ($p < 0.01$). With the jump in R^2 value, this indicates that individual firm traits and decisions explain most changes in program

expenses. On average, when the economy grows and wealth increases, non-profit organizations receive some portion of it and spend it on program expenses. However, if the country's income falls, so do non-profits' incomes and they must downsize.

Looking at the revenue side of the non-profit model and regressions in Table 4 shows a similar progression through the iterative models.

Table 4. Effect of changes in stata GDP on organization revenue

	(1)	(2)	(3)	(4)	(5)
GDP Change %	-0.092***	-0.052***	-0.003	-0.003	0.006***
	(0.001)	(0.002)	(0.002)	(0.002)	(0.001)
Num. Obs.	2,880,054	2,880,054	2,880,054	2,880,054	2,880,054
R2	0.002	0.080	0.082	0.115	0.844
FE: year		X	X	X	X
FE: state			X	X	X
FE: industry				X	X
FE: firm					X

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Note: Regressing a non-profit's total revenue on state real GDP percentage change from prior year to current year, with each current year representing a Form 990 annual filing for tax-exempt status.

Starting at the baseline model, column 1, again suggests a countercyclical relationship. For a one percentage point increase in real GDP, a non-profit's total revenues decrease by 9.2% and this result is significant at the 99% confidence level ($p < 0.01$). Although, this model is without any controls and leaves a large amount of variance unexplained by the chosen explanatory variable. By adding controls for year fixed effects in column 2, state fixed effects in column 3, and industry fixed effects in column 4, more of the variance becomes statistically explained. At the same time, the size of the effect of the coefficient (β_1) decreases. In column 4, controlling for year, region, and industry, shows for a one percentage point increase in real GDP, a non-profit's total revenues decrease by 0.3% but this result is not statistically significant.

By holding firm specific traits as a constant in column 5, the effect flips. For a one percentage point increase in real GDP, a non-profit's total revenue increases by 0.6%, and this

result is statistically significant at the 99% confidence level ($p < 0.01$). The coefficient is a positive value, which tells us that non-profit revenue follows a procyclical trend. Revenues grow when the economy grows but shrink when the economy shrinks.

Table 5 shows the results controlling for individual firm choices by including total revenue as another variable for a budget constrained model. The table compares this budget-controlled model with the firm-fixed effect models from above.

Table 5. Non-profit Expense-side vs. Revenue-side Cyclicality

	(1) **(Program Expenses)**	**(2)** **(Revenues)**	**(3)** **(Budget Controlled)**
GDP Change %	0.006***	0.006***	0.001
	(0.001)	(0.001)	(0.001)
Log(Revenue)			0.823***
			(0.001)
Num. Obs.	2,880,054	2,880,054	2,880,054
R2	0.847	0.844	0.921
FE: year	X	X	X
FE: state	X	X	X
FE: industry	X	X	X
FE: firm	X	X	X

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 5 shows a different explanation for changes in a non-profit's program expenses; total revenue explains most of the variance. For a 1% increase in a non-profits total revenue, their program expenses increase by 0.82%, and this result is statistically significant at the 99% confidence level ($p < 0.01$). Meanwhile, the effect of GDP drops while total fit for the model, indicative in the R^2 value rising by 7.7%, improves. These results imply non-profit finances are elastic. A change in a non-profit's revenue changes their program offerings. If revenue declines, they must reduce expenses for their budget. If revenue remains constant, economic conditions may have only a slight cyclical trend effect, but this result is not statistically significant, and the standard error may bring the effect to zero.

Thus far, the results show that on average all non-profit organizations follow a cyclical trend with the economy that is constrained by their budget, or total available revenues. However, this result may be masking countercyclical behaviors that are invisible in the average effects.

Table 6 shows the results for each quintile of total revenues.

Table 6. GDP Effect on Program Expenses, by Organization Revenue Size (Quintiles)

	Q1 (Smallest)	**Q2 (Small)**	**Q3 (Medium)**	**Q4 (Large)**	**Q5 (Largest)**
GDP Change %	0.009***	-0.002	0.001	0.002	-0.000
	(0.002)	(0.002)	(0.002)	(0.002)	(0.001)
Log(Revenue)	0.861***	0.812***	0.809***	0.765***	0.696***
	(0.001)	(0.001)	(0.001)	(0.002)	(0.002)
Num. Obs.	372,484	389,014	639,112	690,965	788,479
R2	0.946	0.906	0.834	0.861	0.919
FE: year	X	X	X	X	X
FE: state	X	X	X	X	X
FE: industry	X	X	X	X	X
FE: firm	X	X	X	X	X

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Note: Split into quintiles by revenue size according to the base year observation.

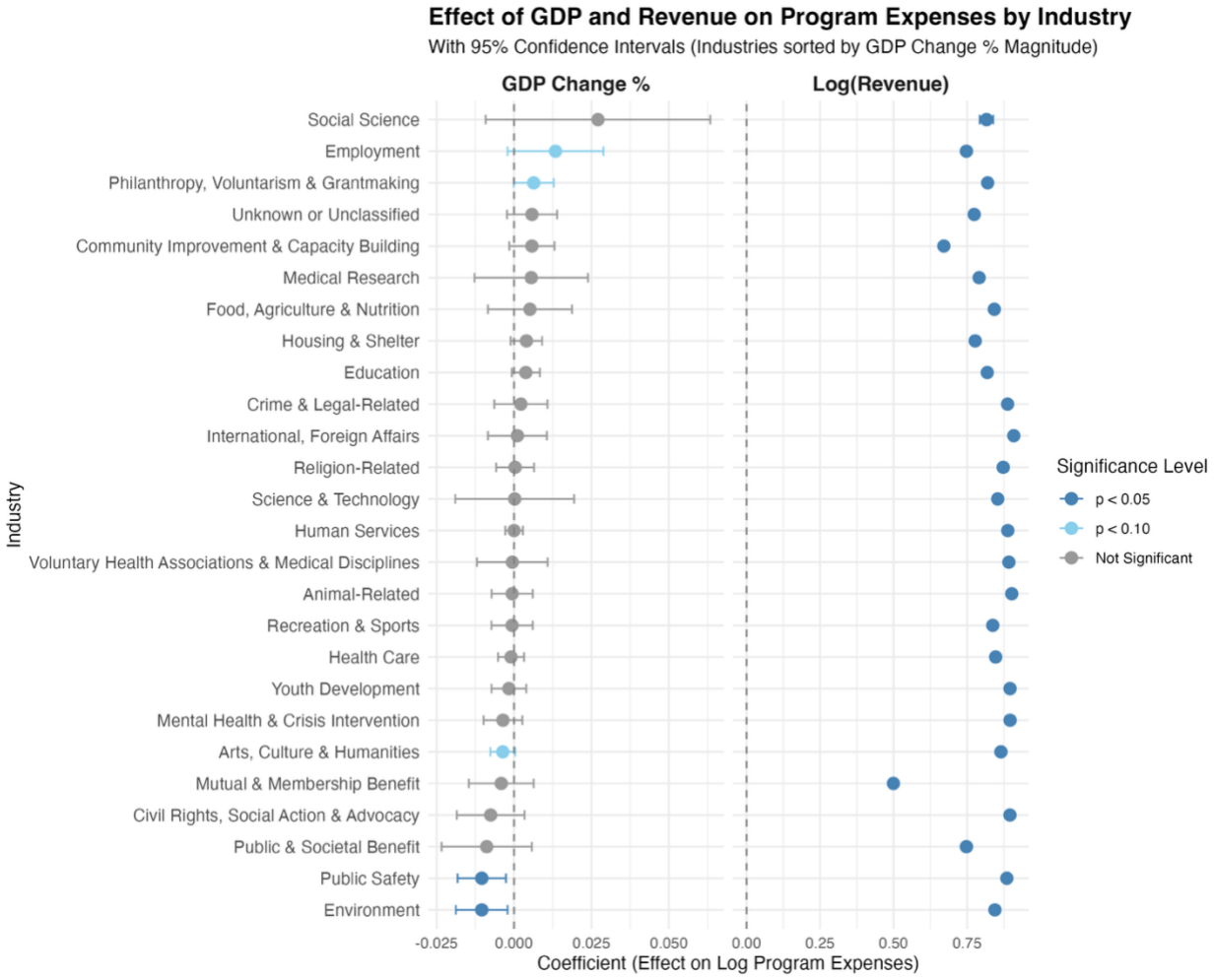
Starting at the results for the 20% smallest organizations, quintile 1 (Q1), shows a strong procyclical trend with economic conditions and available revenues. For a one percentage point increase in real GDP, non-profit program expenses increase by 0.9%. So, if the GDP decreases, so should program expenses, even after controlling for revenue.

Across larger organizations, quintiles (Q) 2-5, show a range in trend. Some may operate countercyclically while others stay procyclical. For a one percentage point increase in real GDP, non-profit's program expenses may decrease by 0.2% ranging to an increase of 0.2%, depending on revenue size. Further, across all sizes the budget constraint is evident. An increase in total revenue is always correlated with an increase in total program expenses; this result is statistically significant and a consistent effect across all organization sizes. As firms get larger, the effect of budget shrinks and the effect of revenue growth decreases from 0.861 to 0.696. Smaller firms

appear to be more budget elastic than larger ones and more affected by changes in the economic conditions of their headquarter state.

Figure 5 (or Table 9 on page 45 in Appendix) summarizes the coefficients when split into each industry and tells a more detailed story than an average regression coefficient.

Figure 5. GDP Effect on Program Expenses by Industry



Across all organizations, there is a statistically significant effect of revenues in determining program expenses.

The GDP effect is only significant for some industries but shows a varying trend overall. Employment and Philanthropy Voluntarism & Grantmaking industries show a procyclical trend. For a one percentage point increase in GDP, program expenses increase by 1.3% and 0.6%,

accordingly, and is significant at the 90% confidence level ($p < 0.1$). In the countercyclical direction, Public Safety and Environment industries have a statistically significant trend at 95% confidence level ($p < 0.05$). For a one percentage point increase in GDP, program expenses also decrease by 1%.

Most other industries show slight or almost zero coefficient with changing GDP. For example, Science & Technology and Human Services industries show exactly zero for the effect of GDP growth on expenses, though this result is not statistically significant. This low significance could be due to the inclusion of many outliers, especially close to zero dollars for many financial variables; or some industries have a low sample size n creating a high standard error relative to the coefficient.

Table 7 shows an altered revenue side model, looking at how different revenue sources change due to economic conditions.

Table 7. Cyclicality of Non-Profit Revenue Sources

	(1) Donations	**(2) Fundraising**	**(3) Govt Grants**	**(4) Membership**	**(5) Investment**	**(6) Other**
GDP Change %	-0.000	0.004***	-0.001	0.002**	0.000	-0.003*
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Num. Obs.	2,880,054	2,880,054	2,880,054	2,880,054	2,880,054	2,880,054
R2	0.754	0.705	0.780	0.859	0.877	0.712
FE: year	X	X	X	X	X	X
FE: state	X	X	X	X	X	X
FE: industry	X	X	X	X	X	X
FE: firm	X	X	X	X	X	X

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The heterogeneity tests show that program expenses are affected by available revenues. The results show some revenue sources follow a procyclical trend, while some are countercyclical, and others have no clear trend. A one percentage point increase in GDP is correlated with a 0.4% increase in Fundraising and 0.2% increase in Membership revenue. These results are significant at 99% confidence level ($p < 0.01$). For the same increase in GDP, Other

revenues decline by .3% on average and is at 99% confidence level ($p < 0.01$). Donations and Government Grants revenue sources have no clear economic trend, both with a coefficient near zero and no statistical significance.

Table 8 shows how each revenue source affected an organization's probability of expanding program expenses during the 2020 Covid-19 recession.

Table 8. Effect of Revenue Sources on Expanding During Covid

	Covid Expansion	**Pre-Covid Robustness Check**
Log(Donations)	0.003***	0.004***
	(0.000)	(0.000)
Log(Fundraising)	0.002***	0.005***
	(0.000)	(0.000)
Log(Govt Grants)	0.011***	0.008***
	(0.000)	(0.000)
Log(Membership)	-0.006***	-0.003***
	(0.000)	(0.000)
Log(Investment)	-0.005***	-0.003***
	(0.000)	(0.000)
Log(Other)	0.000**	-0.000***
	(0.000)	(0.000)
Num. Obs.	2,318,254	2,171,516
R2	0.079	0.058
R2 Adj.	0.079	0.058
FE: year	X	X
FE: state	X	X
FE: industry	X	X

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

In these models, the dependent variable is a Boolean value, so the coefficients represent the change in the probability of expanding programs for every one percent increase in a specific revenue source.

Government Grants showed the strongest correlation. With a tenfold increase in government grant revenues, there is 2.5 p.p. increase in the probability of an organization expanding its programs and this result is highly significant at the 99% confidence level ($p <$

0.01). Donations and Fundraising revenue also show a positive and significant correlation with the probability of expanding programs increasing by 0.7 p.p. and 0.5 p.p., accordingly, given a tenfold increase in the revenue source.

On the other hand, Membership and Investment revenues show a negative correlation with the probability of expansion. For a tenfold increase in Membership Fees or Investment income, the probability of expanding program is correlated with decreasing by 1.4 p.p. and 1.2 p.p., accordingly. These results are also highly significant ($p < 0.01$).

These results are further affirmed by a pre-Covid robustness check of the model, which shows similar correlations across all the revenue sources.

6. Discussion

The iterative macro-models lend some support to altruistic theories about non-profit enterprise, that they expand their services during bad economic times to meet increased community need and scale back during booms, as argued by Pollifroni, Militaru, and Socaciu (2014). However, the remaining the results of this thesis tell a different story.

The initial results with macroeconomic fixed effects (year, state, industry) align with recent research into non-profit macroeconomics by Exley, Lehr, and Terry (2023), who find that non-profit revenues across time, regions, and sectors follow a distinctly procyclical trend. Specifically, during economic downturns, revenues uniformly decline by about .7% for every 1% of income decline. Because this model controls for unobserved, time-invariant firm characteristics, it explains a large degree of variation in the dataset with high statistical significance. By isolating individual firm choices, it provides a highly accurate observation of how firms react to macroeconomic shifts.

The identical, positive coefficients found on both the supply (revenue) and demand (program expense) sides strengthen the conclusion that economic conditions and non-profit finances move in tandem. A change in state GDP has a direct effect on the market for non-profit goods and services, simultaneously impacting the non-profit's maximum supply of capital and its consumer-facing expenditures. This procyclical behavior observed supports DeVides and Harris (2025), who note that as total revenue increases, program expenses also increase. However, their research finds a different magnitude for this relationship, noting that for a 1% increase in revenue, program expenses increase by 0.26% versus a 0.82% increase found in my sample. This difference in magnitude is likely explained by their additional restrictions related to corporate governance and size, as well as my methodology of not splitting types of non-profit expenses.

Clearly, organizations do not blindly adjust their spending precisely in tandem with revenue shocks. As demonstrated by Jones, Kitching, Roberts, and Smith (2012), charity managers are actively engaging in a “spend-save decision” where a marginal change in overall spending is substantially less than the marginal change in revenue. Instead of spending every marginal dollar, nonprofits intentionally smooth their spending across good times and bad times. When revenues increase by 1%, charities save approximately 0.31% of the change to increase liquid assets. Conversely, when revenues decline, charities mitigate deep cuts to program spending by drawing on these past savings. Their findings bring the results of this study in line with the literature. This smoothing behavior, which is most pronounced in organizations with higher savings, shows financial management can moderate revenue constraints during economic shocks.

Examining the effect across different industries provides further support for the budget constraint theory. While some sectors remain insulated from broader state GDP changes, they are

still significantly affected by changes in their specific revenues. This reinforces the principle that an organization's operational capacity is ultimately bound by its available capital.

The results can be distinguished by sources reliant on individual decisions (membership, donations, fundraising) vs external factors (investments, government grants, other). With this perspective, revenue sources depending on a person's willingness to spend or donate (individual decision) are more procyclical, on average, than revenue sources tied to broader institutional decisions (external factors).

An unexpected finding is that direct donations exhibit no clear cyclical trend, which appears to contradict basic economic intuition and prior literature. Bridgeland et al. (2009) observed a distinct decline in donations during the 2008 recession, and theoretically, as aggregate wealth grows, the average household's capacity to donate should follow suit. Furthermore, Rooney et al. (2020) documented a positive trend in overall donations from 2009 to 2020.

This discrepancy in my results may be capturing a different mechanism, though, as the variance in GDP in this thesis is at the state level. It is highly plausible that as a non-profit observes declining wealth within its domestic donor base, it compensates by increasing out-of-state fundraising efforts. If this inter-state marketing is successful, capital from wealthier states may flow in to subsidize the declining state, dampening the procyclical effect. Additionally, because a large share of charitable giving originates from high-net-worth individuals, an opportunity cost dynamic may be at play. During periods of GDP and market growth, the opportunity cost of donating their capital is higher, so donors may prefer to retain wealth for high-yield investments (e.g., stock market) rather than donating. These intuitions may explain the zero change in donations per change in GDP, as it may be observing highly optimized behaviors of individuals or non-profit managers in response to changes in economic conditions.

The probit model measuring correlation between revenue sources and the odds of expanding during the Covid-19 recession is some of the most interesting and policy relevant results. While earlier models were about overall financial behavior, this model compares these revenue practices to a real economic shock, the Covid-19 recession. Although the coefficients are extremely small, this is expectable for a large dataset with a wide range of non-profit types and sizes. Additionally, the Covid-19 recession had unique circumstances that might have made any non-profit expansion more unlikely, since there were mandatory shutdowns across the economy, social distancing requirements, and supply chain issues. But the very high statistical significance and positive results imply these results are still mathematically true correlations about ideal financial practices.

The results highlight a stark contrast between discretionary or market-tied revenues and philanthropic giving. Organizations that may be more reliant on membership fees (e.g., Employment industry) and investment income (e.g., Philanthropy, Voluntarism & Grantmaking) have a negative correlation with program expansion. Economically, this aligns with consumer behavior during a recession: as unemployment spikes, households eliminate discretionary spending, such as club memberships or business association dues, thus starving these specific non-profits of operating cash. Simultaneously, a recession decreases investment yields, forcing organizations with large endowments to reduce or eliminating spending to preserve their capital. This is supported with the findings from Bridgeland et al. (2009), where disposable non-profit spending and foundation philanthropy declined in the Great Recession.

However, the effect of direct donations and fundraising proved uniquely resilient, correlated with positively driving expansion, and contrasting the findings from the Great Recession. Unlike restricted funds or illiquid assets, donations often represent agile, unrestricted

cash flow. This is likely due to the liquidity factor of these revenue sources. Direct donations and fundraising yield unrestricted, liquid cash which can make an organization better able to adapt to the unique circumstances of Covid-19, such as transitioning to virtually provided services, changing work from home policies, and buying personal protective equipment. This liquidity can allow funding expansions immediately, rather than waiting for structural changes or foundation grant policies to respond.

The most effective driver of countercyclical expansion during a recession seems rooted in public policy rather than individual giving. Revenues from government grants had the strongest positive correlation with program expansion. In 2020, massive federal subsidies, specifically the CARES Act, injected unprecedented cash stimulus into the non-profit sector (Havens and Hellmuth 2020). In summary, the results of this probit model suggest that non-profits with pre-existing government grant reliance and connections were uniquely positioned to benefit from this stimulus. Consequently, these organizations were more likely to fulfill the theoretical altruistic, countercyclical theories proposed by Pollifroni, Militaru, and Socaciu (2014). Public policy likely subsidized their budget constraints, improving the ability to respond to the increased demand of Covid-19 for non-profits goods and services.

7. Conclusion

This research sought to explore what specific financial practices non-profit organizations can follow during periods of economic growth that can correlate to capacity to expand during an economic downturn, when a non-profit's services are more needed. To answer this question, I ran longitudinal panel data regression analyses on non-profit tax returns, which offered detailed financial insights of an organization's activities. These regressions quantified the correlation between macroeconomic conditions (State GDP) and the non-profit's financial behavior across

three distinct dimensions: the demand-side of the non-profit market and the effect of macroeconomic conditions in how organizations change their philanthropic programs offered; the supply-side of a non-profit's activities and the effect of macroeconomic conditions on revenue sourcing for an organization; and a revenue-controlled dimension and whether an organization responds to increased demand for philanthropy even under a budget constraint.

By analyzing this comprehensive dataset of U.S. non-profits from 2009 to 2023, this thesis confirms prior literature that the non-profit sector is largely procyclical. An organization's ability to expand its program expenses is constrained by its available revenue, which is constrained by economic conditions. Meaning that as the broader economy contracts, non-profit supply shrinks just as public demand for philanthropy peaks, like during a recession.

However, countercyclical behavior is still possible and observable. The data suggests that organizations can actively engage in financial smoothing, mitigating the severity of revenue shocks to protect program expenditures. Furthermore, the analysis of the Covid-19 recession reveals that improving the chances for expansion, such as to match expanded demand for food services, in a recession is possible, but it is highly dependent on an organization's revenue composition.

Future research should have an even more granular analysis of financial decision making. This thesis was limited to use summarized financial information, instead of analyzing the experiences of non-profit mangers who might prioritize and distribute different sources of revenue, monitor organization-specific financial risks, and rely on other stakeholders to influence program expenditure decisions. Thus, defining specific financial 'practices,' either from interviews to gauge industry 'recommended practices' or testing specific techniques from literature may be a more effective analysis.

Another limitation of this research was not looking into changes in the balance sheet and other accounting decisions. Future research should aim to dissect these mechanics further by examining detailed changes in organizational balance sheets, such as the specific reallocation of assets and liabilities during a shock.

Also, it might be useful to interview non-profit directors, to see if these considerations for withstanding recessions, operating countercyclically, and responding to increased demand of philanthropy during a recession are even active concerns by the leadership. This would vastly strengthen any findings regarding 'best' practices to follow.

Finally, pairing quantitative panel data with qualitative interviews of non-profit directors would provide invaluable context. This could help determine the extent to which countercyclical planning is even an active concern among leadership.

In conclusion, further work should be done to investigate these financial practices to help non-profit enterprise to develop more refined, evidence-based strategies to ensure that non-profits remain resilient precisely when society needs them most.

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Appendix

Appendix 1:

Table 9. Regression results for GDP Effect on Program Expenses, by Industry

	GDP Change %	Log(Revenue)
Recreation & Sports	-0.001	0.837***
	(0.003)	(0.002)
Public & Societal Benefit	-0.009	0.747***
	(0.007)	(0.005)
Mutual & Membership Benefit	-0.004	0.499***
	(0.005)	(0.005)
Community Improvement & Capacity Building	0.006	0.671***
	(0.004)	(0.003)
Arts, Culture & Humanities	-0.004+	0.865***
	(0.002)	(0.001)
Unknown or Unclassified	0.006	0.774***
	(0.004)	(0.003)
Environment	-0.010*	0.844***
	(0.004)	(0.003)
International, Foreign Affairs	0.001	0.908***
	(0.005)	(0.002)
Health Care	-0.001	0.847***
	(0.002)	(0.002)
Education	0.004	0.818***
	(0.002)	(0.001)
Food, Agriculture & Nutrition	0.005	0.842***
	(0.007)	(0.005)
Philanthropy, Voluntarism & Grantmaking	0.006+	0.820***
	(0.003)	(0.002)
Human Services	-0.000	0.888***
	(0.001)	(0.001)
Science & Technology	0.000	0.854***
	(0.010)	(0.006)
Mental Health & Crisis Intervention	-0.004	0.896***
	(0.003)	(0.002)
Religion-Related	0.000	0.872***
	(0.003)	(0.002)
Medical Research	0.006	0.791***
	(0.009)	(0.006)
Youth Development	-0.002	0.896***
	(0.003)	(0.002)
Housing & Shelter	0.004	0.777***
	(0.003)	(0.002)
Animal-Related	-0.001	0.901***
	(0.003)	(0.002)
Voluntary Health Associations & Medical Disciplines	-0.001	0.892***
	(0.006)	(0.003)
Crime & Legal-Related	0.002	0.887***

	(0.004)	(0.003)
Employment	0.013+	0.747***
	(0.008)	(0.006)
Civil Rights, Social Action & Advocacy	-0.008	0.895***
	(0.006)	(0.003)
Public Safety	-0.010**	0.885***
	(0.004)	(0.003)
Social Science	0.027	0.816***
	(0.019)	(0.012)

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Appendix 2:

Link to code repository: https://github.com/jonathantamen/economics-honors-thesis